

Money and Inflation

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Outline

- 1 Introduction: Money
- 2 Money Supply: Basics
- 3 Money Demand
- 4 Inflation: Measurement
- 5 Nominal and Real Interest Rates
- 6 Money Market Equilibrium
- 7 Revenues from Money Creation
- 8 Costs of Inflation



What is Money?



What is Money?

- “Money” is anything that serves as:
 - ▶ Store of Value
 - ▶ Unit of Account
 - ▶ Medium of Exchange
- Money solves the “double coincidence of wants”
- Something is convenient to serve as money if it has several properties:
 - ▶ Hard to be counterfeit
 - ▶ Easy to carry
 - ▶ Be durable
 - ▶ Easily divisible
 - ▶ Be commonly accepted (social convention; laws)
- Different assets satisfy these properties to different degrees: no unique measure of money



What is Money?

Monetary base	M0	M1	M2
Physical Currency Central Bank Reserves	Physical Currency	Physical Currency Demand deposits	Physical Currency Demand deposits Savings deposits Small time deposits Money market mutual funds

- Monetary base: the only part of money supply under direct control of government
- From $M0$ to $M2$: broader concepts, lower liquidity

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Money Supply: Basics

- Central Bank decides how much currency to issue
- But money supply depends also on what commercial banks choose to do
- Typical balance sheet of a commercial bank:

Balance Sheet	
Assets	Liabilities
Reserves	Deposits
Bonds	
Loans	Other Liabilities
Other Assets	Net Worth

- Among the banks' assets (left) we have reserves they keep with the Central Bank (which usually don't earn interest)
- Banks hold reserves:
 - ▶ to meet unexpected withdrawals of deposits
 - ▶ because it is required (regulation)

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Money Supply: Basics

- Most important banks' liabilities (right) is deposits
- But non-deposit borrowing too (long term bonds issued by banks)
- Banks' **net worth** ("patrimônio líquido") = value of assets - value of liabilities

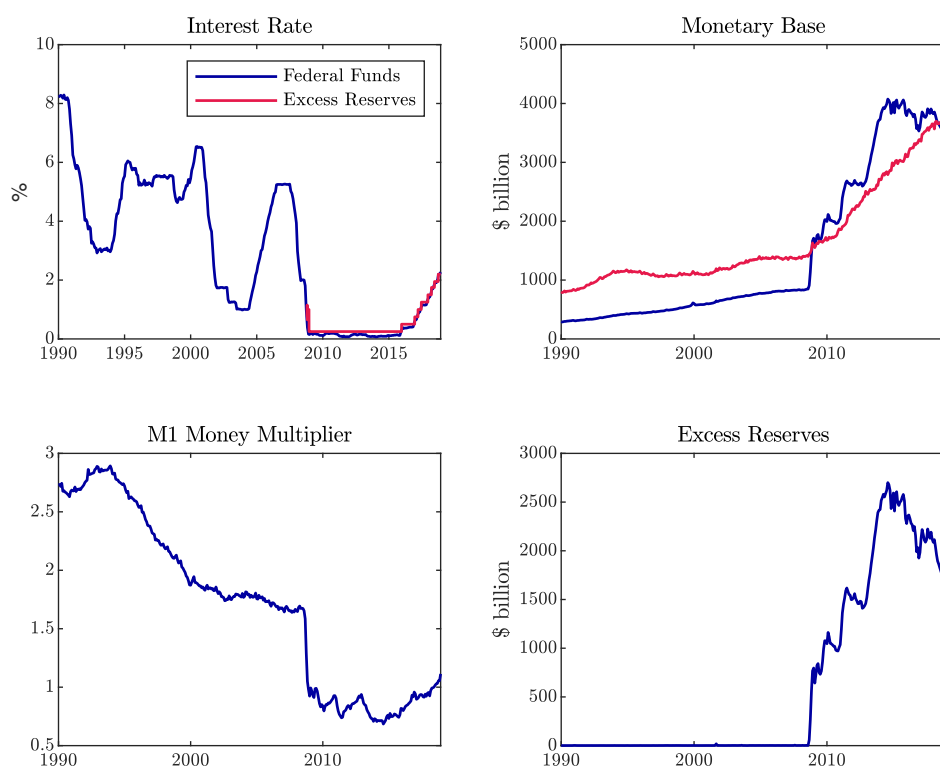
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Changing the Money Supply when Interest Rate is Zero

- The logic of keeping reserves at the minimum required breaks down when interest rate is zero or when CB pays positive interest on reserves
- This implies that changes in monetary base may **not** change M_1 because all new reserves will sit idle in banks and not create new loans and deposits
- This is what happened in the US in the late 2008



Changing the Money Supply when Interest Rate is Zero



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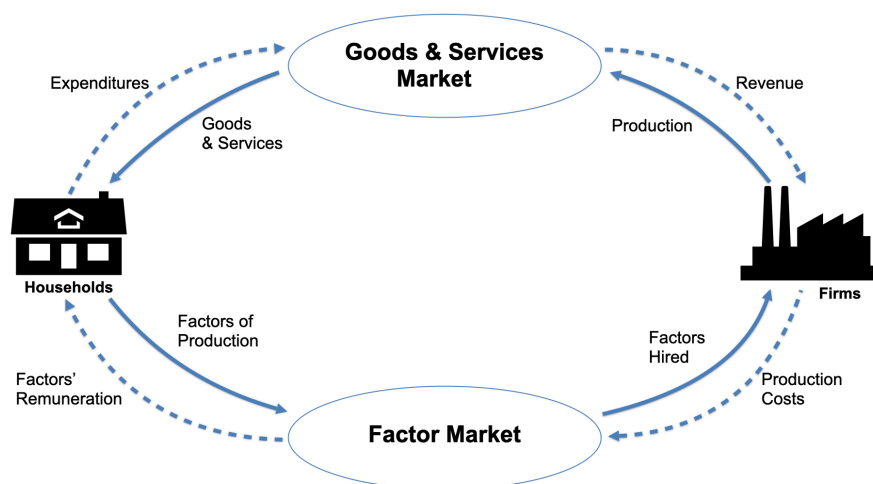


Money Demand

- We focus on M_1 : why do people hold physical currency and checking deposits (with no or very little interest rate)?
- Three reasons (Keynes 1936, Ch. 15):
 - ▶ Transactions motive
 - ▶ Precautionary motive
 - ▶ Speculative motive
- An early theory of the transactions motive: Quantity Theory of Money



Money Demand

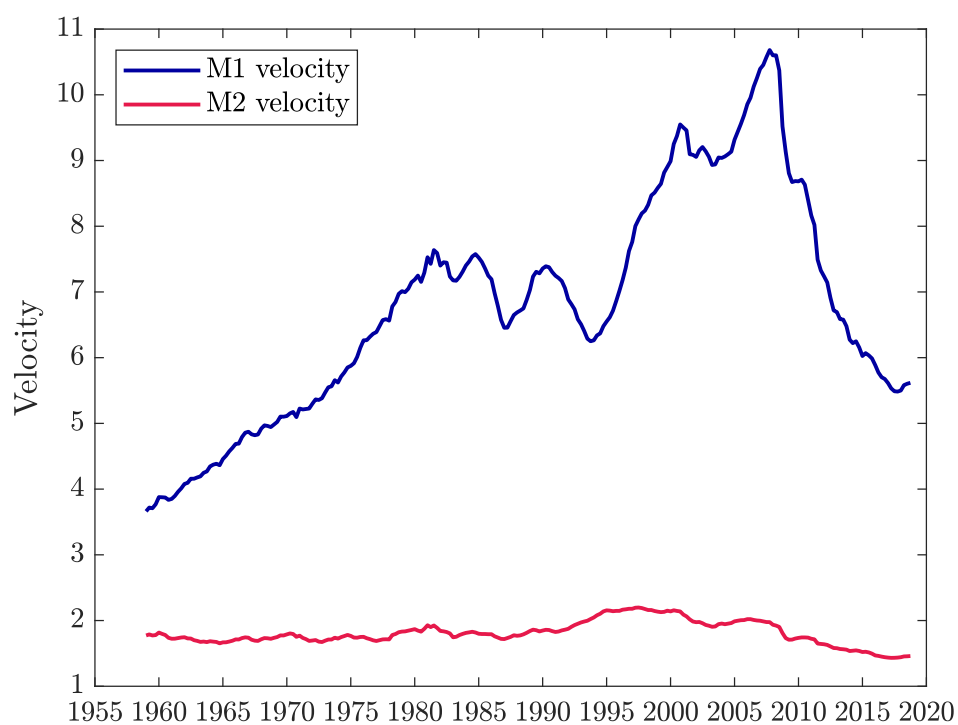


- Monetary flows equal the value of real flows

$$M \cdot V \equiv P \cdot Y$$

- Theoretical arguments make it an *ex ante* theory instead of an *ex post* identity
- This is known as the exchange equation of the QTM

Money Demand



Money Demand

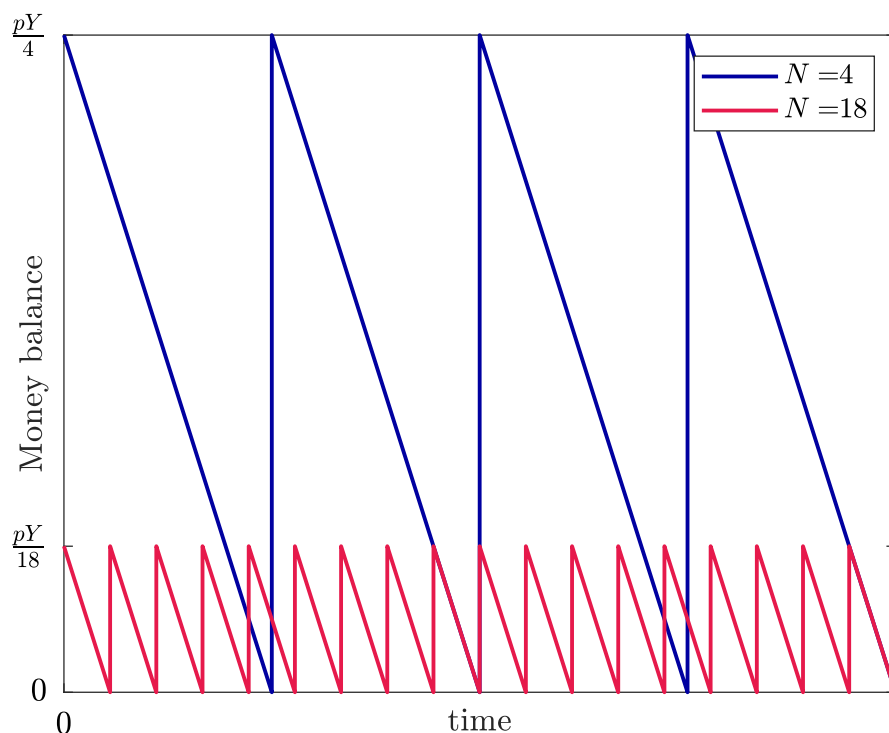
- A more explicit money demand theory comes from the Cambridge Equation:

$$\frac{M^D}{P} = k \cdot Y = \frac{M^S}{P}$$

- Another theory of transactions motive: **the Baumol-Tobin model** (Baumol 1952, Tobin 1956)
 - ▶ There are two types of assets:
 - ★ money (M_1) which pays no interest
 - ★ interest-bearing assets: i
 - ▶ Households don't spend pY at once, but spread it evenly through the time period considered (e.g. one year: $pY/365$)
 - ▶ In order to pay for things, households must use money, and there is a fixed cost F of "going to the bank"
 - ▶ Households decide how many times per period it goes to the bank (N)
 - ▶ Each time households go to the bank they bring their money balance up to $\frac{pY}{N}$, that is then spent linearly until next visit



Money Demand



Money Demand

- The average money balance is:

$$M = \frac{pY}{2N} \quad (1)$$

- Household choose N in order to minimize overall cost of having money for transactions:

$$\text{Min}_N pFN + i \frac{pY}{2N}$$

- F.O.C.:

$$pF - i \frac{pY}{2} N^{-2} = 0$$

- Which can be solved for N :

$$N = \sqrt{\frac{iY}{2F}} \quad (2)$$

- Replacing (2) into (1) we get the average money balance:

$$M = p \frac{Y}{2 \left(\frac{iY}{2F} \right)^{1/2}} = p \sqrt{\frac{YF}{2i}}$$

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Money Demand

- So demand for real balances is:

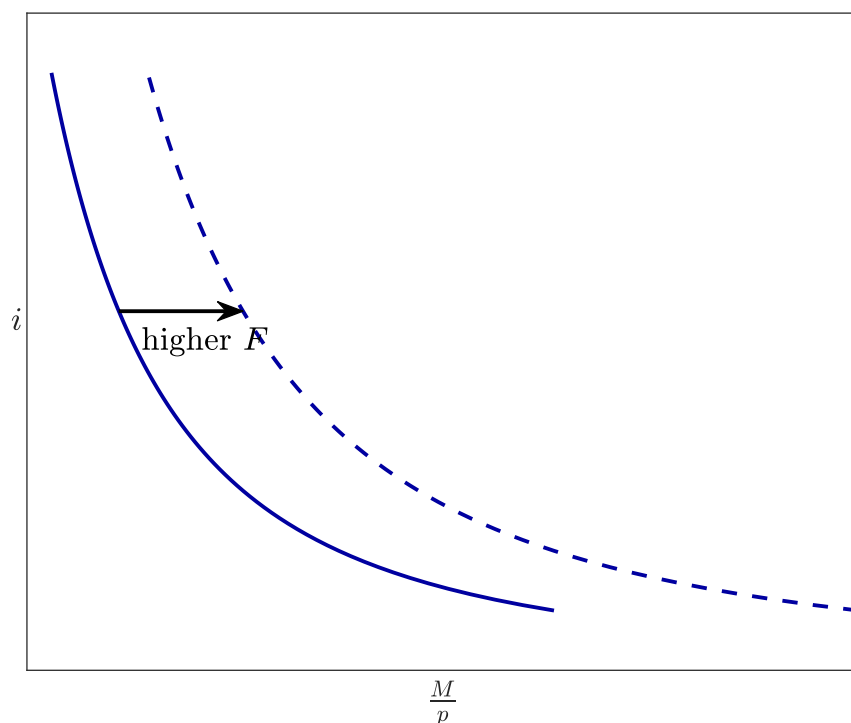
$$\frac{M}{P} = \sqrt{\frac{YF}{2i}} \quad (3)$$

- This money demand depends:

- ▶ Positively on Y
- ▶ Negatively on i
- ▶ Positively on F

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Money Demand



Navigation icons: back, forward, search, etc.

Money Demand

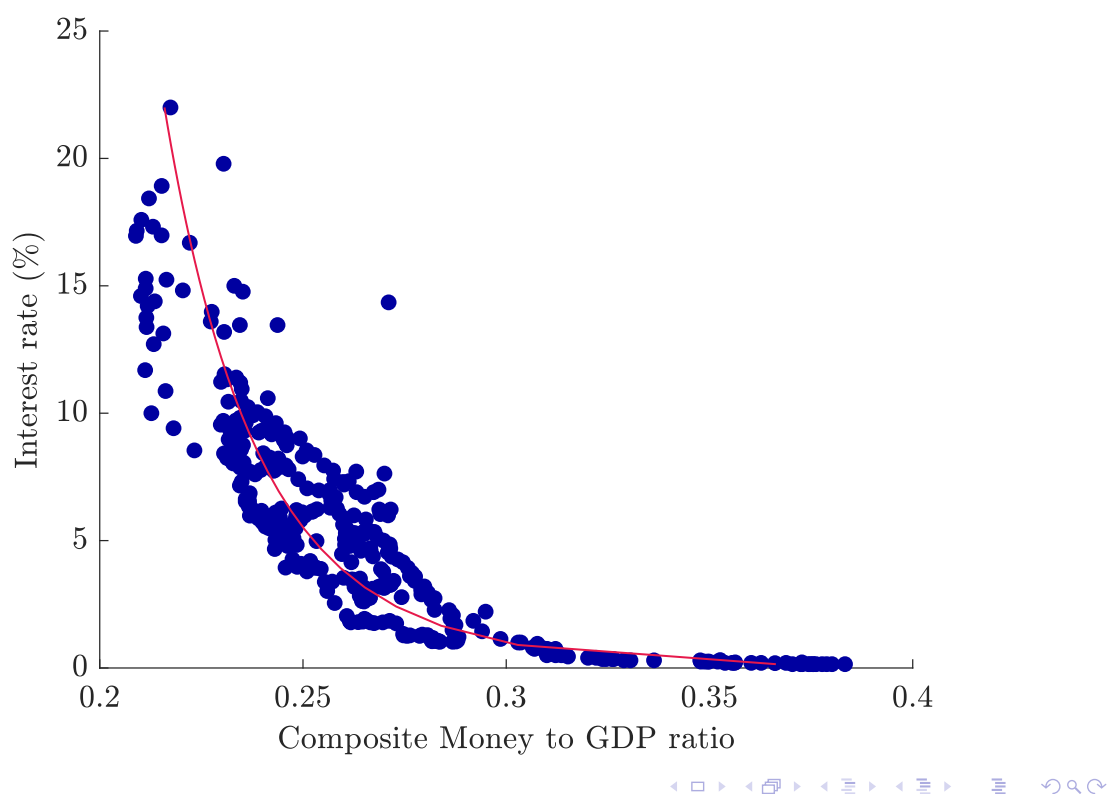
- Although the Baumol-Tobin model makes very specific assumptions:
 - ▶ fixed cost for every trip to the bank
 - ▶ evenly-spread spending
 - ▶ perfect foresight
- We can think that it captures the basic economic forces for money demand, but that the exact formula (3) doesn't hold
- Generalized money-demand function that is increasing in Y and decreasing in i , $m^D(Y, i)$, that has as one special case:

$$m^D(Y, i) = \sqrt{\frac{YF}{2i}}$$

- Is money demand stable?

Navigation icons: back, forward, search, etc.

Money Demand



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Inflation Measurement

- Inflation $\equiv$ generalized increase in the level of prices
- Challenge: define “overall” level of change when prices are changing either in different directions or at different rates
- This is similar to the challenges of measuring real and nominal GDP (see lecture 1)
- We need to define a basket of goods and measure its total price $\Leftrightarrow$ **Price Index**

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Inflation Measurement: GDP Deflator

- The GDP Deflator is a side product of the calculation of real GDP

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

- It weights goods by how much they are produced (their share on GDP)

Example

2017				2018			
Good	Qty	Unit Price	Total	Good	Qty	Unit Price	Total
Wheat	10	50	500	Wheat	11	60	660
Computer	1	1,000	1,000	Computer	2	600	1,200
1,500				1,860			

- real GDP in 2018 at 2017 prices: $11 \times 50 + 2 \times 1000 = \$2,550$

$$\text{GDP Deflator}_{2018} = \frac{1,860}{2,550} \times 100 \approx 73$$

$$\pi_t = \frac{P_t}{P_{t-1}} - 1 \approx -27\%$$

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Measurement: Consumer Price Index

- Consumer Price Indices are more commonly used
- They weight goods by how much they are consumed (survey)

$$CPI_t = \frac{\sum_j q_j p_{jt}}{\sum_j q_j p_{j0}} \times 100$$

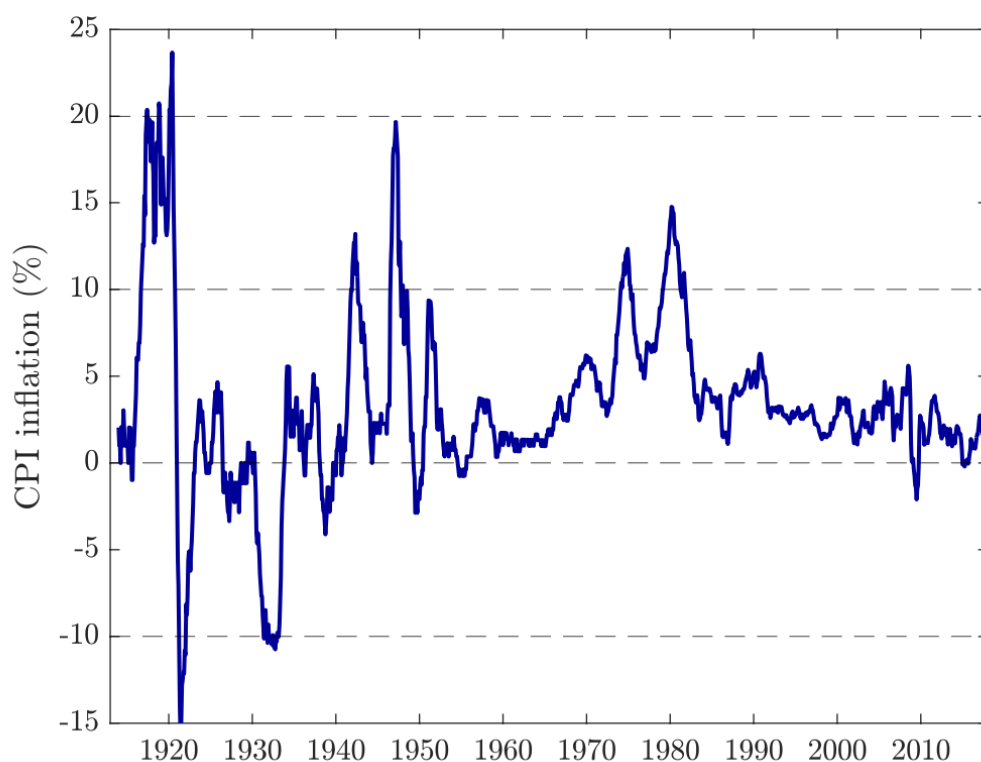
Example

Good (j)	Quantity (q_j)	2017 Price (p_{j0})	2018 Price (p_{jt})
Ferrari (units)	2	100	115
Caviar (Kg)	20	4	3
Champagne (L)	10	2	4
Total Basket		300	330
CPI		100	110

$$\pi_t = \frac{P_t}{P_{t-1}} - 1 = 10\%$$

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Measurement: US Inflation (CPI)



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Nominal and Real Interest Rates

- Typical lending transaction
 - ▶ lender gives a borrower US\$ 1 in period t
 - ▶ borrower pays back US\$ $(1 + i_{t+1})$ in $t + 1$
 - ▶ with i_{t+1} being the interest rate between periods t and $t + 1$
- However, there is no single interest rate for all loans
- We are often interested in expressing interest rates in terms of goods rather than in US\$

Example

In Usuria a one-year loan issued in January 2024 charges an interest rate of 11%. Everyone expects inflation to be 2%. A lender lends US\$ 100 in January 2024. What are they giving up? What do they get in return?

	January 2024	January 2025
Price Index	100	102
Loan issued / repayment received (US\$)	100	111
Loan issued / repayment received (in goods)	$\frac{100}{100} = 1$	$\frac{111}{102} \approx 1.088$

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Nominal and Real Interest Rates

$$1 + r_{t+1} = \frac{\text{Goods afforded at loan repayment}}{\text{Goods afforded at loan issuing}} = \frac{\frac{1+i_{t+1}}{P_{t+1}}}{\frac{1}{P_t}} = \frac{1 + i_{t+1}}{\frac{P_{t+1}}{P_t}} \\ \Rightarrow r_{t+1} \approx i_{t+1} - \pi_{t+1} \quad (4)$$

- Equation (4) is valid for small π_{t+1} and is known as the **Fisher equation**
- It can compute either *ex-ante* or *ex-post* real interest rate

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Money Market Equilibrium

$$M^S = m^D(Y, i) \cdot p \quad (5)$$

- How does money market equilibrium come about?
- If Central Bank increased M^S , equilibrium can be restored through:
 - ▶ increase in p
 - ▶ fall in i
 - ▶ increase in Y
- Understand how each variable responds is extremely important
- **“Classical” View:** money neutrality (real-nominal dichotomy)

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Money Market Equilibrium: Steady State with Constant Money Supply

- Steady State with Y and r constant
- Central Bank holds M^S constant as well
- Conjecture: Then p will be constant as well
- Then i is also constant, which from (5) we get p constant:

$$p = \frac{M^S}{m^D(Y, r)}$$

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Money Market Equilibrium: Steady State with Growing Money Supply

- Steady State with Y and r constant
- Now Central Bank increases M^S at constant rate, μ : $M_{t+1}^S = (1 + \mu)M_t^S$
- Conjecture: Then π will be constant at μ
- If $\pi_{t+1} \equiv \frac{p_{t+1}}{p_t} - 1 = \mu$, then:

$$i_{t+1} = r + \pi_{t+1} = r + \mu$$

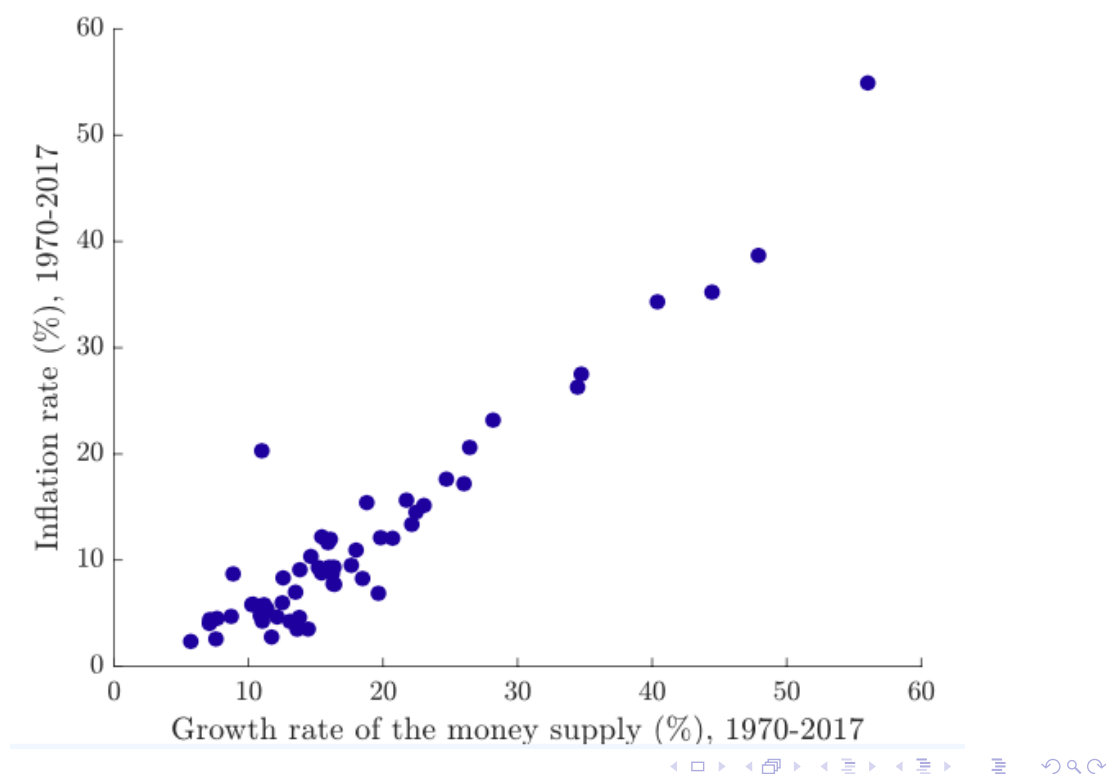
- Then money market equilibrium in period t :

$$\begin{aligned} M_t^S &= m^D(Y, r + \mu)p_t \\ \Rightarrow M_t^S(1 + \mu) &= m^D(Y, r + \mu)p_t(1 + \mu) \\ \Rightarrow M_{t+1}^S &= m^D(Y, r + \mu)p_{t+1} \end{aligned}$$

- which implies money market equilibrium in $t + 1$, confirming the conjecture

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Money Market Equilibrium: Steady State with Growing Money Supply



Money Market Equilibrium: Steady State in Growing Economy

- Steady State with Y growing at constant rate g and constant r
- Central Bank increases M^S at constant rate, μ : $M_{t+1}^S = (1 + \mu)M_t^S$
- To find π differentiate (5) with respect to time

$$\frac{dM^S}{dt} = \left[\frac{\partial m^D(Y, i)}{\partial Y} \frac{dY}{dt} + \frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt} \right] \cdot p + m^D(Y, i) \frac{dp}{dt}$$

- Now divide each side by (5):

$$\begin{aligned} \frac{\frac{dM^S}{dt}}{M^S} &= \left[\left(\frac{\partial m^D(Y, i)}{\partial Y} \frac{Y}{m^D(Y, i)} \right) \frac{\frac{dY}{dt}}{Y} + \frac{\frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt}}{m^D(Y, i)} \right] + \frac{\frac{dp}{dt}}{p} \\ \mu &= \left[\left(\frac{\partial m^D(Y, i)}{\partial Y} \frac{Y}{m^D(Y, i)} \right) g + \frac{\frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt}}{m^D(Y, i)} \right] + \pi \end{aligned}$$

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Money Market Equilibrium: Steady State in Growing Economy

- Let the elasticity of money demand with respect to GDP be $\eta \equiv \frac{\partial m^D(Y, i)}{\partial Y} \frac{Y}{m^D(Y, i)}$.
- Assume a money demand with constant elasticity; thus:

$$\mu = \left[\eta g + \frac{\frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt}}{m^D(Y, i)} \right] + \pi$$

- If inflation is constant, then $i = r + \pi$ is also constant:

$$\mu = \eta g + \pi$$

- Or $\pi = \mu - \eta g$, which tells that inflation is indeed constant
- Economy with higher growth rate and higher η imply lower inflation

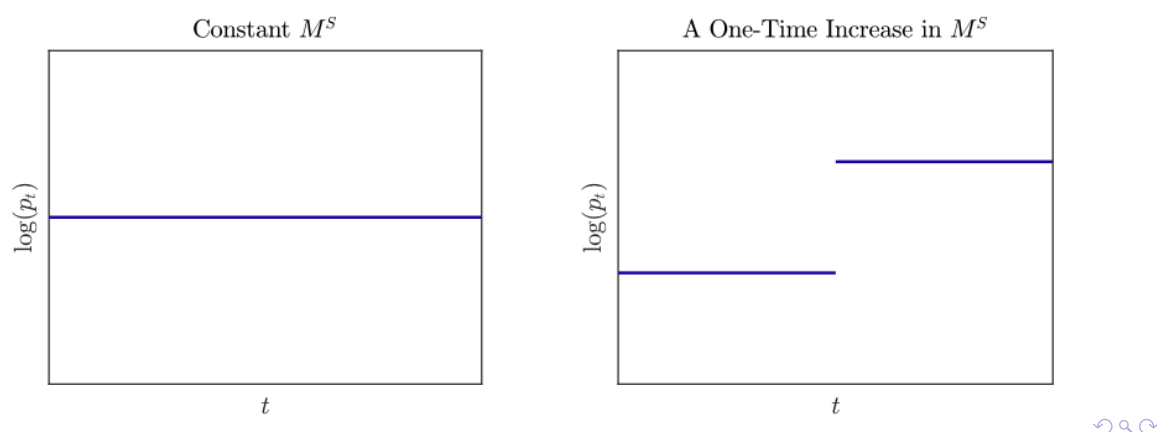
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Money Market Equilibrium: Steady State in No-Growth Economy

One-Time Increase in M^S (case of constant M^S)

- Sudden, unexpected and permanent increase in money supply, from M^S to $M^{S'}$.
- Before this change and after it, $p = \frac{M^S}{m^D(Y,r)}$
- Jump in p :

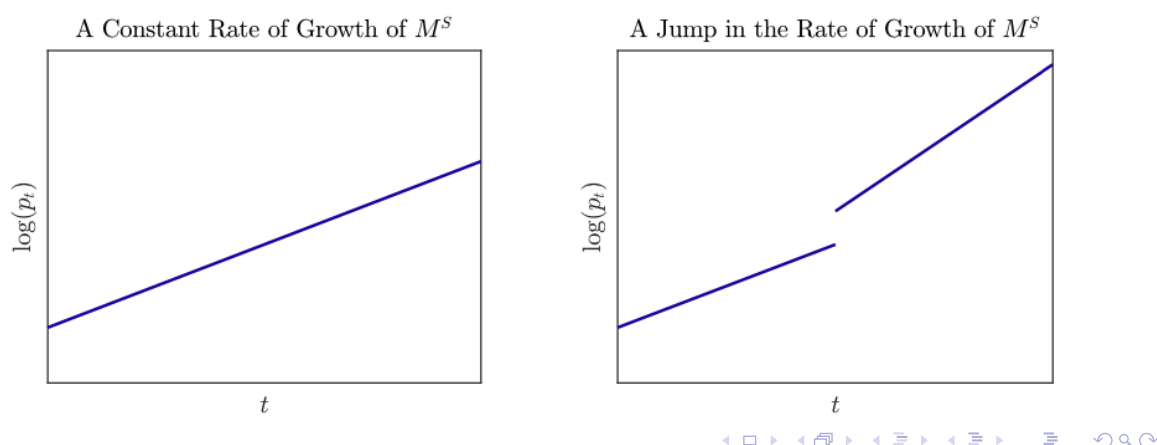
$$\frac{p'}{p} = \frac{M^{S'}}{M^S}$$



Money Market Equilibrium: Steady State in No-Growth Economy

One-Time Increase in Money Supply Growth Rate

- Sudden, unexpected and permanent increase in money supply growth rate, from μ to μ' .
- The guess that inflation rises from $\pi = \mu$ to $\pi' = \mu'$ is incomplete as it implies i changing from $r + \pi$ to $i' = r + \pi'$, which would reduce m^D
- But M^S doesn't change when g increases; then p must rise to clear money market



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Seignorage

- Nowadays most countries keep inflation quite low, though not zero
- One reason for having positive (and high) inflation rate is *seignorage*
- Increase in monetary base is a way for government get a loan that it will never have to repay and that doesn't pay interest
- Government Budget Constraint:

$$B_{t+1} + p_t \tau_t + [M_{t+1}^B - M_t^B] = p_t G_t + (1 + i_t) B_t \quad (6)$$

- This can be rearranged as:

$$M_{t+1}^B + B_{t+1} = p_t [G_t - \tau_t] + (1 + i_t) B_t + M_t^B$$

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Seignorage

- Who does the government tax when it increases the monetary base?
- Anyone holding money is implicitly paying taxes when monetary base increases
- Increase in money supply increase prices, which reduces the purchasing power of money
- This is why seignorage revenue is also called “inflation tax”
- Limit: the tax base (m^D) reduces with higher inflation rates (thus higher nominal interest rate)

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The Cost of Inflation

- Why is inflation generally seen as undesirable?
- One reason comes from the Baumol-Tobin money demand model
- Higher inflation brings higher nominal interest rate, which increases the number of times people go to the bank (each costing F)
- Total cost of trips to the bank:

$$\begin{aligned}\text{Cost} &= NF = \sqrt{\frac{iYF}{2}} \\ &= \sqrt{\frac{(r + \pi)YF}{2}}\end{aligned}\tag{7}$$

- Milton Friedman thus advocated having $i = 0$, known as the “Friedman Rule”
- This calls for a deflation rate $\pi = -r$

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The Cost of Inflation

- Why is inflation generally seen as undesirable?
- Another reason: **menu costs** (real resources needed to implement price change)
- Minimizing menu costs require $\pi = 0$ (instead of deflation)
- Notice that $\pi = 0$ does **not** eliminate menu costs, as individual prices would still move up and down
- Another cost of inflation: **relative price uncertainty** (people don't look at all the prices and relies on the prices of few different goods to get a sense of how much one US\$ is worth; this affects consumption and production decisions)
- Another cost: seignorage raised by **private banks** (bank deposits) –high inflation leads to excessive entry into banking industry

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